Heart of Algebra

Section 1: Multi-Step Linear Equations & Identity Conditions

1 If $5(2x - 1) - 3(x + 4) = 4(x - 3) + 7$, what is the value of x?

a. 4

b. 6

c. 8

d. 12

2 For what value of a will the equation $3(ax - 2) + 14 = 12x + 8$ have infinitely many solutions for all real numbers x?

a. 2

b. 4

c. 6

d. 8

3 If $\frac{2x - 5}{3} = \frac{x + 4}{2}$, what is the value of $3x + 1$?

a. 22

b. 64

c. 67

d. 70

Section 2: Linear Inequalities & Feasible Regions

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4 Solve the compound inequality for x: $-5 < 2x + 3 \leq 13$.

a. $-4 < x \leq 5$

b. $-4 \leq x < 5$

c. $-1 < x \leq 8$

d. $x > -4$

5 Which of the following points (x, y) is a solution to the system $y > 2x - 3$ and $y \leq -x + 4$?

a. $(1, 2)$

b. $(3, 5)$

c. $(0, -4)$

d. $(2, 4)$

6 If $-3x + 7 \geq 22$, what is the greatest possible value of x?

a. -5

b. -3

c. 5

d. 15

Section 3: Systems of Linear Equations

7 In the system of equations $4x - 3y = 11$ and $2x + 3y = 7$, what is the value of $x - y$?

a. 1

b. 2

c. 3

d. 4

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8 For what value of k does the system $kx + 4y = 8$ and $3x + 2y = 5$ have no solution?

a. 3

b. 6

c. -6

d. 12

9 A store manager buys x pens at 1.50 each and y notebooks at 4.00 each, spending a total of 78.00. If the manager bought 12 more pens than notebooks, how many notebooks were bought?

a. 8

b. 10

c. 12

d. 15

Section 4: Rate, Unit Conversions & Real-World Modeling

10 A car uses 8 liters of fuel per 100 kilometers. How many liters of fuel are required for a trip of 350 kilometers?

a. 24

b. 26

c. 28

d. 32

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11 A technician charges a flat dispatch fee of 45 plus 30 per hour of repair. If the total bill was 195, how many hours did the repair take?

a. 4

b. 5

c. 6

d. 7

12 Pump A drains a water reservoir in 6 hours, and Pump B drains the same reservoir in 3 hours. Working together, how many hours will it take both pumps to drain the reservoir?

a. 1.5

b. 2.0

c. 2.5

d. 4.5

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